Customer Churn Prediction and Retention Analysis

End-to-End Machine Learning and Business Retention Insights

Executive Summary

This project examined whether customer information can support earlier, capacity-aware retention decisions. The analysis covered 1,200 subscription customers and followed a controlled workflow from data audit through cleaning, exploratory analysis, feature engineering, model development, final evaluation, explainability, and a retrospective top-20% retention simulation.

Logistic Regression, Random Forest, and Histogram Gradient Boosting were compared using leakage-safe five-fold stratified cross-validation. The tuned Random Forest achieved the strongest training-validation Average Precision at 0.9782 and was retained. Sigmoid calibration was selected using training-only evidence, and the final model achieved held-out ROC-AUC 0.8994 and Average Precision 0.9819.

At the locked probability threshold, precision was 0.9899 and recall was 0.4734. For a campaign with a hard 20% contact limit, deterministic ranking selected 48 of 240 held-out customers; all 48 historically churned. This produced precision@20% 1.0000, recall@20% 0.2319, and lift 1.1594 versus random selection. The result supports risk prioritization, but it does not show that contacting a customer prevents churn.

1,200	0.8994	0.9819	1.1594
customers	held-out ROC-AUC	held-out Average Precision	lift at top 20%

Business Problem and Objectives

Customer churn reduces recurring value and creates replacement costs, while retention teams have finite capacity. The project objectives were to build a reproducible churn-risk model, identify the strongest measured risk signals, explain individual predictions, and assess whether ranking can concentrate historical churners within a 20% contact budget. The project deliberately separates prediction from intervention effectiveness.

End-to-End Analytical Workflow

1 Prepare	2 Develop	3 Explain	4 Prioritize
Audit, clean, split, and prevent leakage	Compare, tune, calibrate, and test once	Assess global behavior and anonymized cases	Rank risk, apply capacity, and test scenarios

Decision takeaway

The principal business takeaway is to use the model as a prioritization aid rather than an automated retention decision. Current customers should be scored, ranked, and filtered through a fixed contact capacity, eligibility rules, privacy controls, and a controlled intervention pilot.

Evidence Boundary

Supported use	Not established by this study
Prioritize limited outreach using validated risk ranking and human review	Causal churn reduction, realized profit, production readiness, or automated action

Dataset and Data Preparation

Dataset Overview

The source contains 1,200 customers and 20 columns: one identifier, one binary churn target, and 18 predictors. Churn is unusually prevalent: 1,036 customers (86.33%) have label 1 and 164 (13.67%) have label 0. The high prevalence means accuracy alone is not an adequate measure of usefulness.

Four numerical fields contain missing values: satisfaction score 48, monthly usage 36, monthly charges 30, and total charges 24. In total, 138 cells across 132 customers are missing. There are no complete duplicate rows and no duplicate customer identifiers. Statistical outliers and unusual service combinations were retained because the available evidence did not show that they were erroneous.

Measure	Verified result	Decision relevance
Dataset	1,200 rows x 20 columns	18 predictors plus ID and target
Churn distribution	1,036 churn / 164 retained	86.33% positive class
Missingness	138 cells across 132 rows	Training-fold median imputation
Duplicates	0 rows; 0 repeated IDs	No records removed
Fixed split	960 train / 240 test	Stratified, random state 42

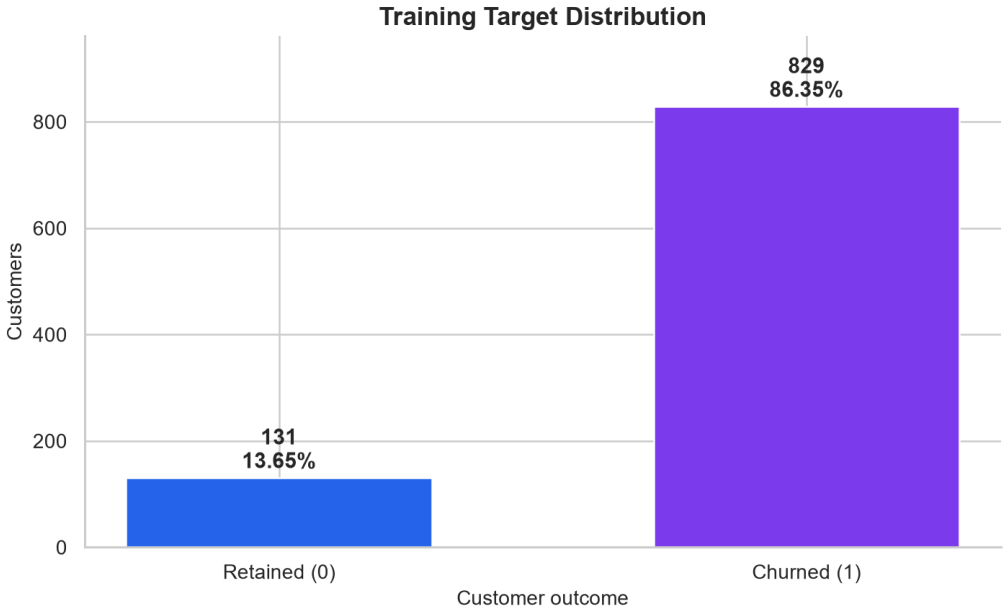


Figure 1. Training target distribution; the high churn prevalence makes accuracy alone misleading.

Preparation and Leakage Controls

Cleaning standardized representation only; it changed no customer values and excluded no rows. Missing numerical values were preserved in the cleaned file and median-imputed inside training-fitted pipelines. Categorical values use an explicit missing category and one-hot encoding with safe handling for unseen levels. Numerical predictors are scaled after imputation.

The data was split once into 960 training rows and 240 final-test rows using an 80/20 stratified split with random state 42. Customer ID and churn were explicitly excluded from predictors. Feature engineering, imputation, scaling, encoding, calibration, and model fitting were all contained inside leakage-safe scikit-learn pipelines. The held-out test partition remained protected until final evaluation.

Exploratory Analysis and Feature Engineering

Observed Churn Associations

Training-only EDA found the clearest separation in monthly usage. Median usage was 70.75 GB among churned customers and 28.00 GB among retained customers. Churn rose from 57.45% in the lowest usage quintile to approximately 99.5% in the upper two quintiles, a nonlinear relationship that requires confirmation that usage is measured before the prediction cutoff.

Month-to-month customers had 91.10% observed churn, compared with 80.00% for one-year and 79.08% for two-year contracts. Satisfaction was lower among churned customers, while two or more late payments and three or more complaints coincided with elevated churn. Tenure and monthly charges showed non-monotonic patterns, and payment method provided relatively weak separation. These are predictive associations, not causal findings.

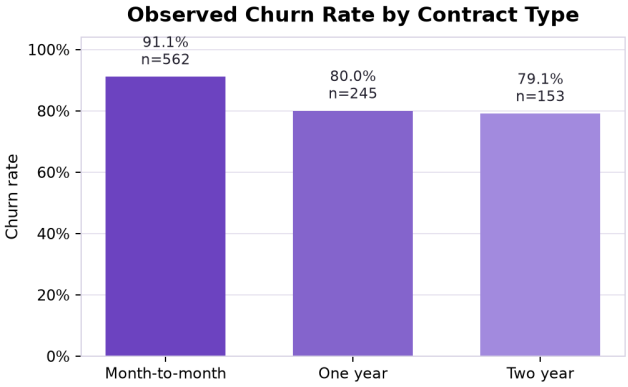


Figure 2. Training-only contract association; group counts are shown above each bar.

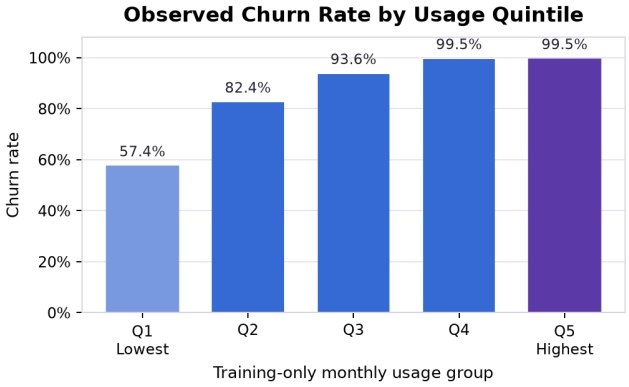


Figure 3. Training-only usage association; missing usage values are excluded from quintiles.

Engineered Features

Seven deterministic features were added without using target values: tenure in years, complaints per active month, a payment-friction score, complaint-satisfaction interaction, month-to-month contract by tenure, late-payment by monthly-charge interaction, and usage per charge. These features express duration, intensity, and combined customer-friction signals. They preserve missingness for training-fold imputation and do not establish that any representation improved outcomes by itself.

Feature	Purpose
tenure_years	Relationship duration in an interpretable scale
complaints_per_active_month	Complaint intensity relative to active exposure
payment_risk_score	Combined recent complaint and late-payment friction
complaint_satisfaction_interaction	Service issues occurring with lower satisfaction
monthly_contract_tenure	Contract-by-tenure interaction observed in EDA
late_payment_charge_interaction	Payment difficulty combined with current charge
usage_per_charge	Usage intensity relative to monthly charge

All associations and derived features are predictive representations, not causal treatment effects.

Machine Learning Development and Evaluation

Training-Only Model Comparison

Logistic Regression, Random Forest, and Histogram Gradient Boosting were evaluated alongside a naive prior reference. All models used the same five stratified folds and complete feature/preprocessing pipeline. Sixteen hyperparameter configurations were evaluated for each main model using Average Precision as the primary selection metric.

The tuned Random Forest produced mean cross-validation Average Precision 0.9782 +/- 0.0060 and ROC-AUC 0.8822. Tuned Logistic Regression achieved Average Precision 0.9752, while tuned Histogram Gradient Boosting achieved 0.9741. These differences are small: Random Forest exceeded Logistic Regression by 0.0030 and Histogram Gradient Boosting by 0.0041 in mean Average Precision.

Random Forest was selected before final-test access because it combined the highest mean Average Precision and ROC-AUC, joint-best precision@20% of 0.9949, and relatively low fold-to-fold Average Precision variation. Its 400-tree ensemble is more complex than Logistic Regression, so the simpler model remains a reasonable fallback where transparency or operating cost outweighs the modest ranking gain.

Tuned model	CV AP	AP SD	CV ROC-AUC	Precision@20%
Logistic Regression	0.9752	0.0151	0.8816	0.9897
Random Forest	0.9782	0.0060	0.8822	0.9949
Histogram Gradient Boosting	0.9741	0.0054	0.8715	0.9949

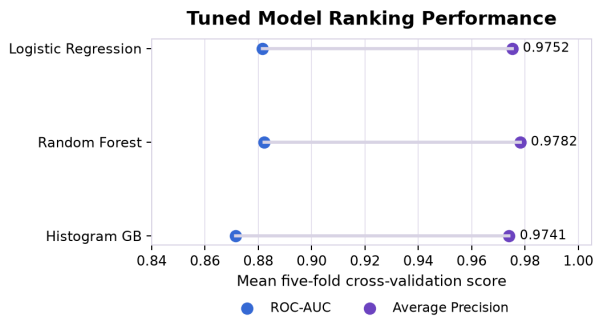


Figure 4. Tuned ranking scores are close; Random Forest has the highest mean AP.

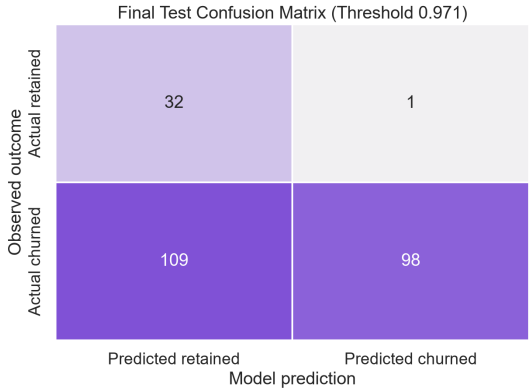


Figure 5. Locked-threshold held-out outcomes: 98 TP, 32 TN, 1 FP, and 109 FN.

Locked Final Evaluation

Sigmoid calibration was selected using nested training-only folds. It reduced OOF Brier score from 0.1061 to 0.0840, log loss from 0.3328 to 0.2752, and ECE from 0.1181 to 0.0219 without material ranking loss. Held-out Brier score was 0.0825 and log loss was 0.2627; calibration is improved, not perfect.

Accuracy	Precision	Recall	F1	ROC-AUC	Average Precision
0.5417	0.9899	0.4734	0.6405	0.8994	0.9819

Why accuracy is low
Accuracy 0.5417 is below the 0.8625 always-churn baseline. The high cutoff produces precision 0.9899 but misses 109 churners, so high precision does not imply high recall. ROC-AUC 0.8994 and AP 0.9819 instead support useful cross-threshold ranking.

Threshold is not capacity
The locked threshold 0.970709 selected 99 of 240 customers (41.25%). A real 20% campaign must rank all scores and select exactly 48 using the declared tie-breaker. The threshold was not retuned after test access.

Model Explainability and Customer Insights

Global Model Behavior

Original-feature permutation importance measured the decrease in held-out Average Precision after each input was shuffled. Monthly usage was the dominant feature with mean importance 0.1072 +/- 0.0164. Contract type ranked second at 0.0044, followed by satisfaction score at 0.0036 and late payments at 0.0017. The remaining features were close to zero relative to repeated-permutation variability.

Observed predictions rose sharply with usage and generally fell as satisfaction increased. Month-to-month customers had higher average fitted risk than fixed-term customers. Importance and grouped predictions describe model behavior and association; they do not prove that changing a feature will change churn.

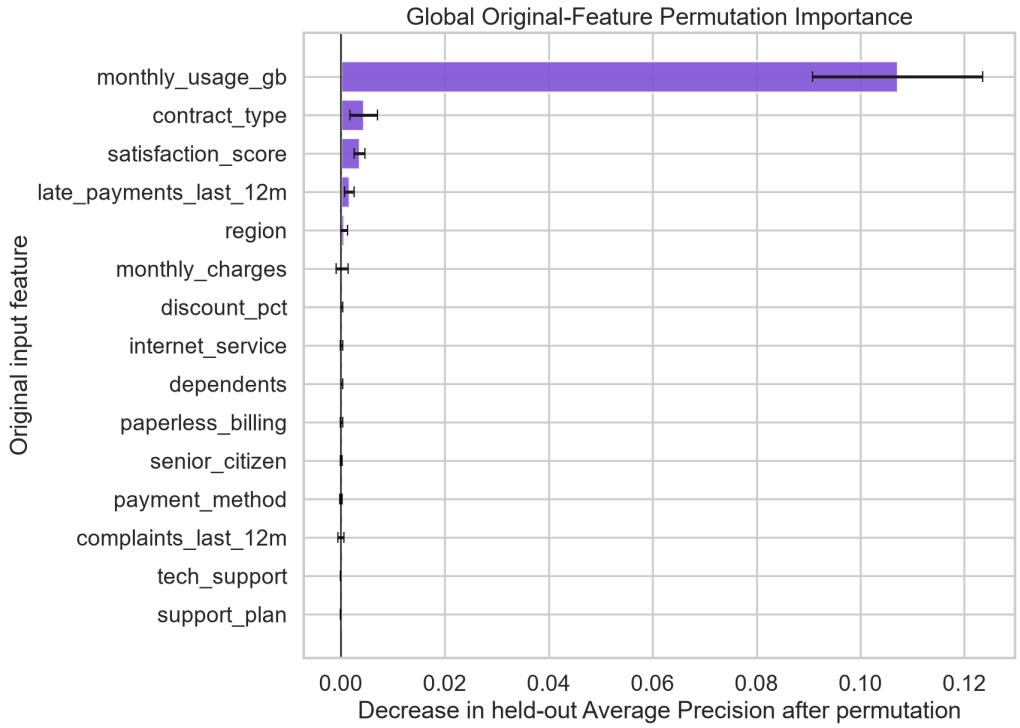


Figure 6. Held-out original-feature permutation importance measured by Average Precision decrease.

Anonymized Local Explanations

Local explanations used one-feature-at-a-time probability sensitivity against training-derived reference values. Customer A was a correct churn prediction at probability 0.9783; usage of 75.8 GB was the dominant risk-raising local signal. Customer B was a correct non-churn prediction at 0.6423; a one-year contract and usage of 35.4 GB strongly lowered fitted risk relative to their references, despite opposing age and payment signals.

Customer C was a false negative at probability 0.9706, only 0.000073 below the locked threshold. Usage of 76.3 GB raised fitted risk, while the one-year contract and charge effect lowered it slightly. The case illustrates the fragility of binary decisions near a high threshold. Local effects are non-additive and non-causal.

Case	Probability	Actual	Predicted	Result	Main local factors
Customer A	0.9783	Churn	Churn	Correct	Usage raised risk most; small charge increase; tenure/satisfaction slightly lowered risk.
Customer B	0.6423	Retained	Retained	Correct	One-year contract and low usage lowered risk; age and late payments raised it.
Customer C	0.9706	Churn	Retained	False negative	Usage raised risk; one-year contract and charge effects kept score just below cutoff.

Explanations describe fitted-model sensitivity and association; they do not establish causation.

Retention Simulation and Business Impact

Top-20% Capacity Policy

The campaign policy is separate from threshold-based classification. It ranks the unchanged held-out probabilities descending and selects exactly floor(N x 0.20) customers. Original held-out order is the predeclared secondary key for ties and does not use customer identity or actual churn. For 240 evaluated customers, 48 were selected; the locked threshold itself would have selected 99 and therefore cannot enforce the contact limit.

Locked classification rule	Operational campaign rule
Probability >= 0.970709; selected 99/240 (41.25%); retained for honest evaluation	Rank probability descending; select exactly 48/240 (20%); stable original-order tie-break

All 48 selected customers historically churned, giving precision@20% 1.0000 and recall/capture@20% 0.2319. With overall churn prevalence 0.8625, lift was 1.1594. Uniform random selection with the same contact count would capture 41.4 churners in expectation, so model ranking concentrated 6.6 additional historical churners. "Captured" means present in the retrospective group, not successfully retained.

Evaluated	Selected	Churners selected	Precision@20%	Recall@20%	Lift	Random capture
240	48 (20.00%)	48	1.0000	0.2319	1.1594	41.4

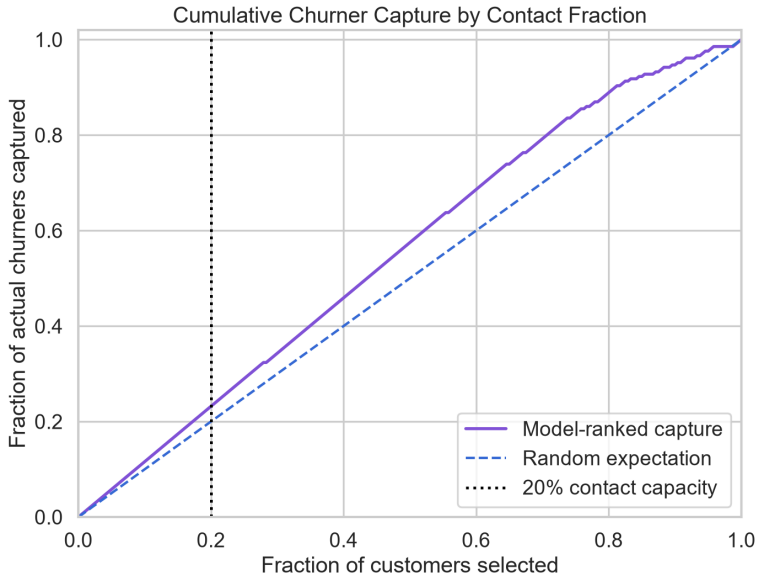


Figure 7. Solid purple shows model-ranked capture; the dashed blue diagonal is random selection, and the vertical dotted line marks the 20% contact limit.

Hypothetical Financial Scenario

The financial illustration assumes a $5 contact cost, $150 value per successfully retained would-be churner, and 10% intervention effectiveness. These are hypothetical USD accounting inputs, not company figures or measured treatment effects. Under these assumptions, model ranking costs $240, yields 4.8 expected retained customers, $720 gross value, $480 net benefit, and 200% scenario ROI.

Random selection under the same budget produces $381 expected net benefit and 158.75% scenario ROI. The incremental expected model benefit is $99. At the base cost and value, model net benefit is $120, $480, and $1,200 at assumed success rates of 5%, 10%, and 20%. The model strategy is positive in 23 of 27 tested cost-value-effectiveness scenarios, with a base break-even success rate of 3.33%.

Assumptions	Model-ranked estimate	Random-selection estimate
$5/contact; $150/retained; 10% success	$240 cost; $720 gross	$240 cost; $621 gross
Expected net benefit	$480; ROI 200.00%	$381; ROI 158.75%
Incremental model benefit	$99 versus random	-

All financial values are illustrative scenario estimates, not realized company profit or measured causal uplift.

Recommendations, Limitations and Conclusion

Evidence-Linked Retention Actions

First, use service-quality and plan-fit reviews for eligible high-risk, high-usage customers because usage is the strongest model signal. Second, prioritize proactive support and complaint resolution for eligible high-risk customers with low satisfaction or unresolved issues. Third, offer compliant, voluntary payment-support outreach where recent late payments indicate possible friction. Fourth, provide contract and renewal assistance to eligible month-to-month customers when a suitable plan exists.

These actions are intervention hypotheses. They can create agent cost, discount leakage, privacy concerns, or a poor customer experience. A randomized holdout is necessary to estimate incremental retention uplift and margin before broader rollout.

Observed support	Proposed action	Mechanism	Main uncertainty
Usage is the strongest signal	Service-quality and plan-fit review	Resolve reliability or mismatch	Usage timing and non-causality
Lower satisfaction and complaints	Proactive support resolution	Reduce service friction	Agent cost; outreach can backfire
Late-payment history	Optional payment support	Reduce involuntary loss	Privacy, compliance, and fairness
Month-to-month contract association	Renewal assistance	Offer suitable voluntary options	Discount leakage and self-selection

Implementation Roadmap

Prepare and score	Pilot and learn
1. Validate current, consented inputs	5. Review eligibility, privacy, and fairness
2. Generate probabilities with the saved model	6. Assign approved interventions
3. Rank customers by churn probability	7. Run a controlled pilot and record outcomes
4. Apply the strict top-20% contact limit	8. Monitor drift, calibration, capture, cost, and subgroups

Limitations

The dataset is modest and has unusually high churn prevalence. Missing values and unresolved service/usage relationships remain. Predictor timing is not documented well enough to eliminate all temporal-leakage risk. The final evaluation uses one held-out sample rather than external or temporal validation, and only 33 retained customers appear in the test set. Calibration and the very high threshold may shift in a future population.

Model importance and local sensitivity are associative rather than causal. No treatment-response data or verified company cost and value inputs exist, so the financial analysis is hypothetical. Customer scoring also requires purpose limitation, access control, fairness review, human oversight, and monitoring for subgroup harm. The project is not production-ready without those controls and a causal pilot evaluation.

Conclusion

Recommended next decision

The project demonstrates that a calibrated Random Forest can rank churn risk effectively and concentrate historical churners within a limited contact budget. The strongest result is operational prioritization, not proof of churn prevention. The appropriate next step is a governed randomized retention pilot that measures incremental outcomes and replaces illustrative economics with verified business inputs.